To Engage or Not to Engage? Social Activities and Survival Beliefs in Older Adults*

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Abstract

While survival belief is central to major life decisions, its social determinants remain understudied. Social engagement is proven to influence various health outcomes, yet the direction of these relationships stays controversial, with quasi-experimental designs being laid aside. It is also unexplored what happens once individuals withdraw from established social life. To address these gaps, this paper leverages the potential randomness in the timing of individual (dis)engagement behavior to examine the causal effects of social (dis)engagement on the subjective probability of unsuccessful survival (SPUS) of young-old Chinese, using a panel of 11,412 respondents from the 2011–2020 China Health and Retirement Longitudinal Study. A dynamic difference-in-differences approach, with various constructs of treatment and control groups, was employed to validate the estimation. Several robustness checks, including factor analysis and propensity score matching, were further conducted. The primary findings are that social engagement reduces older people's SPUS by up to 2.80 percentage points immediately and by 6.67 percentage points one period later. Conversely, social disengagement shows no immediate effects, but its post-treatment impact can reach up to 14.76 percentage points among subgroups. Interestingly, the engagement effects persist only among non-switchers, fading once switching begins, and the disengagement effects also emerge only among non-switchers. In sum, this study provides partial evidence of stronger disengagement effects, though these effects occur neither instantly nor universally.

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1 Introduction

Humans are social animals. People come to understand the world and themselves, including their future, by interacting with the outside and learning from others' experiences. Not surprisingly, individuals update their survival beliefs across time when a series of life events happen to or are observed by them (e.g., Post and Hanewald, 2013; Costa-Font and Vilaplana-Prieto, 2022; Sloan, 2024). It is well-documented that people have estimates about their own lifespan and leverage such estimates, whatever consciously or unconsciously, in their major life decisions, such as retirement planning, financial decisions, and labor force participation (e.g., Griffin et al., 2012; Mittal et al., 2020; Lou and Carr, 2022). Despite heterogeneity in perceived risk of dying across, typically, age cohorts and gender (e.g., Payne et al., 2013; Chen et al., 2021), the judgment of personal longevity is found, in general, a powerful predictor of actual mortality, especially for middle-aged and older people (Jylhä, 2011), as it is constructed upon similar risk factors, including medical conditions, family history, health behaviors, and subjective well-being, that predict actual survival rates (e.g., Kotter-Grühn et al., 2010; Ronconi et al., 2012; Griffin et al., 2013; Kim and Kim, 2017; van Solinge and Henkens, 2018; Becchetti et al., 2023).[1]

While the analysis of subjective survival/mortality is abundant in how it impacts individual economic decisions, in its demographic, medical, health behavioral, and environmental drivers, e.g., epidemic or diagnosed diseases, smoking behaviors, alcohol consumption, road traffic, climate, and air quality, as well as in its co-evolution with life tables (e.g., Hurd and McGarry, 2002; Lundborg and Andersson, 2008; Andersson, 2011; Mussio et al., 2024; Brown and Pepper, 2024), its social determinants are fairly understudied. However, the role of social factors in shaping both physical health and subjective well-being has been proved just as important as health behaviors (Haslam et al., 2018), and garnered increasing

[1] Sloan (2024) provides a systematic review of the accuracy of subjective beliefs, where various influential studies conclude that the discrepancy between subjective survival and objective mortality is small, with Perozek (2008) even showing that the former outperforms life tables in predicting future mortality rates.

attention (Marmot, 2015; Xue et al., 2020).

Social engagement has been proven to exert an influence on health, though not always positively (e.g., Kabayama et al., 2017; Harling et al., 2020; Nakagomi et al., 2023), and a deprivation of it can be more detrimental than living alone to older people's mental health (e.g., Djundeva et al., 2019; Oshio and Kan, 2019). Specifically, the measure of social engagement spans empirical research from paid work, volunteering, formal participation in organizations like clubs, associations, or community groups, to leisure activities. And health outcomes it explains encompass self-reported health (SRH), alcohol use, depressive symptoms, five-year survival rates, specific health shocks (e.g., heart attack, cardiovascular disease and COVID-19), cognitive functioning, life satisfaction, and happiness, among others (see Morita et al., 2010; Kelly et al., 2017; Miao et al., 2019; Rodgers et al., 2019; Cheng et al., 2024). Such studies usually adopt the broader definition of social capital (e.g., Nahapiet and Ghoshal, 1998; Putnam, 2000; Lin, 2002) and link their findings to two key concepts within this framework: social cohesion and social networks.[2]

Nevertheless, it is unfortunate that, to date, very little research has kept an eye on relationships between social engagement and survival beliefs, whether measured in a "live-to" or "die-by" framework. Equally important is the further question of what will happen to older people once they withdraw from their established social life. None of the existing empirical literature has looked into the inverse scenario where individuals shift from participating in social activities toward detaching from them, which is deemed a natural process of aging (Cumming and Henry, 1961), though not for all. However, if a shift toward engagement triggers a positive change in health perception, withdrawal is likely to cause a negative change. This negative change is likely to be even larger if individuals are more sensitive to the disappearance of social ties and other benefits potentially derived from social activities

[2]Social cohesion refers to the connectedness and solidarity among members of a group, often described as a group attribute manifested through mutual trust, shared social norms, and reciprocity. Social networks, on the other hand, are typically understood as the web of relationships among individuals or groups, through which resources, information, and support are facilitated and exchanged.

than to the creation of them, a phenomenon widely observed across numerous contexts where negative experiences often have longer-lasting and more intense consequences than positive ones (Baumeister et al., 2001).

On top of them, previous studies with a focus on social engagement cast more attention to productive, organizational, or formal voluntary participation, these events are, however, supposed to be not equally applicable to all the societies, considering some cultures or institutions tend to favor more informal participation (Wang et al., 2019). Furthermore, interactions with family members, friends and neighbors predominate in the literature when looking into leisure activities, while a more comprehensive notion of social activities, including emerging behaviors, such as virtual chats and exercise-oriented social occasions like square dancing which is very popular among older Chinese is still quite scarce. However, recent studies show that the use of social media platforms can affect people's health, particularly their subjective well-being, through the dynamics of social capital (e.g., Sabatini and Sarracino, 2017; Castellacci and Tveito, 2018). In this study, we incorporate eight different social activities, including online chats and exercise-oriented social behaviors.

Similarly, the related aging literature heavily situates itself in developed countries, such as the US, EU, and Japan, as these societies have experienced population aging for a longer period. However, developing nations have already started to confront aging challenges, with around 80% of older people expected to live in low- and middle-income countries by 2050, and unfortunately, the buffer time they have to adapt to an aging society is much shorter (World Health Organization, 2021). In particular, China is among the cohort facing the fastest aging growth and anticipated to have over a third of the population to be 60 years old or more by 2050, similar to today's Japan.

More importantly, the vast majority of the health literature on social participation with observational data barely touches upon quasi-experimental research designs. However, the direction of the relationships remains controversial, with research pointing out that healthier

individuals are more likely to engage in social activities and vice versa (Xue et al., 2020). Thus, developing a more reliable setting is necessary to further decode this puzzle.[3]

Given these, the primary objective of this paper is to identify the causal effects of social engagement vis-à-vis disengagement, summarizing eight types of social activities, on the subjective probability of unsuccessful survival (SPUS) of young-old Chinese. Following Fadlon and Nielsen (2019), it leverages the potential randomness in the timing of social (dis)engagement, based on a panel of 11,412 respondents from the 2011–2020 China Health and Retirement Longitudinal Study, to create a difference-in-differences (DiD) design. It employs a dynamic approach to assess the parallel trends assumption, conditional on a set of time-varying covariates, and constructs alternative treatment and control groups, using individuals who engaged only on an infrequent basis and those who did not yet engage but would do so later, respectively.

Moreover, it applies Callaway and Sant'Anna's (2021) (henceforth, CS) DiD with multiple periods estimator, designed specifically for staggered DiD setups, to allow for covariate-specific pre-trends and to further address potential heterogeneity in treatment timing and effects. Four additional robustness checks have been conducted to further corroborate the results. Specifically, we create an alternative treatment variable based on factor analysis to explore the influence of social capital, which also increases the number of observations in the analysis; present multilevel ordered logistic regressions to account for the ordinal nature of our dependent variable (DV); assess whether our estimates are robust to a wide range of smooth convex and concave monotonic transformations of the DV; and combine propensity score matching (PSM) with DiD to enhance the comparability between treatment and control groups. To conclude, this study is among the first to use a quasi-experimental design to examine how social behaviors interplay with health, and to the best of our knowledge, the first to focus specifically on survival beliefs. Likewise, it is also the first to explore the disengagement scenario.

[3]Although the relationship might be reciprocal, it is important to understand which effect comes first.

As suggested in previous studies, subjective survival synthesizes information considered in typical SRH and beyond. For instance, Kutlu-Koc and Kalwij (2017) demonstrate that individuals possess private information even after controlling for general health and common health behaviors; potential genetic and unexposed health risks, revealed through family history (e.g., Zick et al., 2014; van Solinge and Henkens, 2018; Donnelly et al., 2020), as well as disparities in health literacy and access to medical resources associated with educational levels and socioeconomic status (e.g., Jylhä, 2011; Arpino et al., 2022), are generally not factored into the evaluation of current health status but found to shape survival expectations. That being said, the intention is not to criticize the use of SRH, but rather to provide an insightful complement to the relevant strand of literature.

In sum, our primary findings indicate that: (a) social engagement lowers older people's SPUS by as much as 2.80 percentage points (6.2% of the sample mean) immediately after engagement begins, and by 6.67 percentage points (14.7% of the sample mean) one period later; (b) social disengagement shows no immediate effects, but it can elevate the SPUS by as much as 14.76 percentage points (32.6% of the sample mean) one period later, suggesting a gradual erosion of social ties following the withdrawal. When separating respondents into those with continued (dis)engagement and those who subsequently switch out of treatment, we find that the engagement effects among switchers fade out once switching begins, and they do not exhibit any disengagement effects. Further, retirement status does not appear to influence the effects of either engagement or disengagement on SPUS. However, in the case of disengagement, no increase in the SPUS is observed among those whose spouses are simultaneously socially inactive.

The remainder of the paper is organized as follows. The next section introduces the data. Section 3 discusses the empirical strategies. Section 4 presents the main results. Section 5 are robustness checks. Discussions and conclusions are drawn in Section 6.

2 Data Description

The panel assembled for this study is built upon the China Health and Retirement Longitudinal Study (CHARLS), a nationally-representative survey launched by the Institute of Social Science Survey, Peking University. Centering on individuals aged 45 and over, CHARLS is modeled on the U.S. Health and Retirement Study to provide a comprehensive reflection of the middle-aged and older population in China. Data were gathered in 2011, 2013, 2015, 2018, and 2020, respectively, at the individual, household, and community level, from 28 provinces and 150 counties and districts, with an attrition rate of 12–14% between each wave. Descriptive statistics of all the variables used in this study are reported in Table SI1 (see Supporting Information).

We base our analysis on a panel containing 11,412 young-old Chinese aged between 55 and 69, a critical stage for the vast majority of Chinese to undergo lifestyle transitions and adapt to retirement life.[4] The rationale for this choice is threefold: a) our DV is age cohort-specific, b) people in their 70s and 80s are much less likely to join a new social activity and much more likely to drop out of their existing ones, and c) a body of literature indicates that age 70 marks a divide between underestimation and overestimation of remaining life duration (Sloan, 2024).

2.1 Subjective Probability of Unsuccessful Survival

Aligned with previous work on subjective survival, our measure is derived from a question in the CHARLS survey and rescaled to a range of 0 to 1 for interpretation (see, e.g., Hurd and McGarry, 2002; Hurd, 2009; Post and Hanewald, 2013; Boyer et al., 2020; O'Dea and Sturrock, 2023). Specifically, respondents are asked to rate their likelihood of living to age 75 (for those aged 55–64) or age 80 (for those aged 65–69). The responses are classified

[4] In general, the retirement age for Chinese women spans from 50 to 55, while for Chinese men, it is 60 to 65.

into five levels: "almost certain," "very likely," "maybe," "not very likely," and "almost impossible." We code these responses on a scale where higher values indicate higher perceived probabilities of unsuccessful survival. Thus, a value of 1 represents "almost impossible" (highest subjective probabilities of unsuccessful survival) and 0 represents "almost certain" (lowest subjective probabilities of unsuccessful survival). Research has shown that people in their 50s and 60s tend to be relatively pessimistic about their survival prospects (e.g., Wu et al., 2015; O'Dea and Sturrock, 2023; Apicella and De Giorgi, 2024). Adjusting this subjective bias in survival probabilities can endue older people with better preparation for health issues in later stages.

2.2 Social Activities

We take into account eight social activities in total: 1. visiting or talking to friends, 2. playing board games like Mahjong or chess, 3. providing help to family, friends, or neighbors who do not live with the respondent, 4. joining exercise-oriented social occasions, such as sports and dance, 5. participating in community-related organizations, 6. doing voluntary or charity work, 7. attending educational or training courses, and 8. using the internet for chatting purposes. These activities also apply to the definition of spousal activities. If individuals did not participate in any of these activities previously but do so in the current waves, their treatment status will be coded as 1. Conversely, in the disengagement setting, the treatment status will be coded as 1 if respondents participated in at least one activity previously but do not in the current waves.

2.3 Time-Varying Controls

In addition to individual age and retirement status, we include some time-varying control variables to account for three essential health factors: serious and chronic diseases, disabilities, and functional abilities related to Activities of Daily Living (ADLs) and Instrumental

Activities of Daily Living (IADLs). Individuals experiencing deterioration in any of these factors may withdraw from established social activities while simultaneously increasing their survival risks. In contrast, relief or recovery from these conditions may encourage participation in social activities and reduce the subjective probability of unsuccessful survival.

Table 1: Questions for diseases, disabilities, and functional abilities

	Description
Diseases	
Answer: Yes=1 or No=0	
Questions 1–14	Whether you have been diagnosed with: 1. hypertension, 2. dyslipidemia, 3. diabetes or high blood sugar, 4. cancer or malignant tumor (excluding minor skin cancers), 5. chronic lung diseases (such as chronic bronchitis and emphysema), 6. liver disease (except for fatty liver), 7. heart attack (such as coronary heart disease, angina, congestive heart failure, and other heart problems), 8. stroke (including cerebral infarction and hemorrhage), 9. kidney disease, 10. stomach or other digestive diseases, 11. emotional, nervous, or psychiatric problems, 12. memory-related disease (such as dementia, brain atrophy, and Parkinson's disease), 13. arthritis or rheumatism, or 14. asthma.
Disabilities	
Answer: Yes=1 or No=0	
Questions 1–5	Do you have: physical disabilities, brain damage or intellectual disability, vision problem, hearing problem, or speech impediment.
ADLs and IADLs	
Answer: Don't have any difficulty=1, have difficulty but can still do it=2, have difficulty and need help=3, cannot do it=4	
Questions 1–11	Do you have any difficulty with: dressing, bathing, eating, getting into or out of bed, using the toilet, controlling urination and defecation, doing household chores, preparing hot meals, shopping for groceries, managing your own money, or taking medications.

Source: CHARLS 2011–2020.

Table 1 summarizes all relevant questions from CHARLS used to construct the corresponding variables. To incorporate all the information, we create two variables, counting the number of newly diagnosed serious or chronic diseases and number of disabilities, re-

spectively. We then conduct a principal component analysis (PCA) based on polychoric correlation matrices to address multicollinearity by extracting and converting essential features of functional ability variables (ADLs and IADLs) into two major, orthogonal components, which explain 82.34% of the total variance. Importantly, social engagement or disengagement is unlikely to have a direct impact on critical illness, disability, or loss of independent living ability. Therefore, these four variables are consistently allowed to vary throughout the panel.

2.4 Controls with Pre-treatment Levels

In addition to the aforementioned time-varying covariates, we control for important individual characteristics related to survival perceptions: SRH, mental health, life satisfaction, body pain, weekly working hours, annual net income, drinking behavior, monthly outpatient service utilization in the last year, and hospitalization history in the last year. Since these variables are potentially influenced by social engagement or disengagement (for instance, participation in social activities may lead to reduced working hours, enhanced life satisfaction, or, in the case of joining sports or dance clubs, an increased likelihood of body pain), raising concerns about bad controls,[5] they are allowed to vary only across pre-treatment survey waves and will be fixed at their values one period prior to the onset of treatment. Nevertheless, to further assess the robustness of our estimates, we also provide comparisons where these covariates are allowed to fully vary over time.

[5] Angrist and Pischke (2009) explain "bad controls" in the context of identifying the causal impact of college graduation on earnings. Although attending college likely influences occupation, controlling for occupation can introduce selection bias, even if college attendance were randomly assigned. However, empirical studies find that simply ignoring such variables also comes with costs (e.g., Brand, 2006; Barnette et al., 2021), and sometimes, is insufficient to validate the parallel trends assumption, particularly when there are time-varying confounders are not included (Roth et al., 2023). For further discussion on why essential time-varying characteristics should not be ignored and the potential advantages of using pre-treatment covariates, see Caetano et al. (2022) and Roth et al. (2023).

2.5 Individual Characteristics Comparisons

It may seem unusual that some individuals are never socially engaged. Although they might have participated very occasionally in certain social activities, consistently reporting to CHARLS that they have not socialized with people outside their households indicates very rare social behavior. Given this, comparisons between the treated and control groups in terms of essential individual characteristics may provide some insights into this matter.

In the engagement setting, our panel includes 7,937 respondents in the treated group and 3,475 respondents in the control group. Among them, more males are observed in the treated group than in the control group (50.9% versus 48.4%), while differences in the number of diseases, ethnicity, religion, working hours, and life satisfaction are negligible. The respondents in the treated group are more likely to be younger (average age 61 in the treated group versus 62 in the control group), to have completed retirement formalities (13.6% versus 7.7%), to be more educated (6.2% fewer illiterate individuals and 7.8% more with a middle or high school degree), to live in downtown areas (17.9% versus 11.4%), to earn higher incomes (4,633 Yuan versus 3,133 Yuan), to be non-disabled (86.4% versus 82.1%), to be better at living independently (0.3 points lower in the treated group for both components of ADLs and IADLs; higher scores indicate greater difficulties), to rate their health lower (77.4% versus 79.1% for fair or better), yet to report better mental health (0.29 versus 0.32; higher scores indicate worse mental health), and to have fewer children (14% versus 11% for a single child, compared to 11.9% versus 14.9% for four children). Similarly, there are 0.8% more married respondents and 1.1% fewer never-married respondents in the treated group compared to the control group.

It is also insightful to compare differences in individual characteristics between respondents who shift toward disengagement and those who consistently participate in social activities. In this regard, our panel includes 5,778 respondents in the treated group and 5,179 respondents in the control group. Among them, more males are observed in the

treated group than in the control group as well (53.1% versus 48.5%), while differences in the number of diseases, age, religion, marital status, and life satisfaction are negligible. The respondents in the treated group are more likely to have more children (13.6% versus 22.8% for a single child, compared to 14.5% versus 8.4% for four children), to not have completed retirement formalities (13.6% versus 27.8%), to be less educated (14% more illiterate individuals and 14.3% fewer with a middle or high school degree), to reside outside downtown areas (14.8% versus 31%), to be disabled (14.3% versus 9.5%), to be slightly worse at living independently (0.14 and 0.21 points higher in the treated group for the two components of ADLs and IADLs), to work longer per week (24.7 hours versus 19 hours), to have lower income (3,035 Yuan versus 6,421 Yuan), to rate their health higher (77.5% versus 73.4% for fair or better), yet to report worse mental health (0.29 versus 0.24). Similarly, there are 3.1% more ethnic minorities in the treated group compared to the control group. These comparisons between the treated and control groups in both settings are also summarized in Table SI2 and Table SI3.

3 Empirical Strategy

3.1 Difference-in-Differences Design

As much as policy shocks seen in many DiD studies are not random but associated with regional characteristics, older people who engage in social activities could be systematically different from those who do not. However, the particular timing of engagement behavior is likely unpredictable given that the questionnaire only looks at its occurrence one month ahead of survey dates. Suppose a person participated in an activity in June but did not in July and is surveyed in August. They will be recorded as not engaging in social activities. In fact, a plethora of respondents reported engagement in one wave (e.g., 2013), but not in the subsequent wave (e.g., 2015), and then again in later waves (e.g., 2018), suggesting

an irregular behavior across time. Therefore, following Fadlon and Nielsen (2019), where the timing of health shocks to individuals is potentially random, and counterfactuals are households that experience the same shock but a few years in the future, we utilize the potential randomness in the timing of social engagement to create a quasi-experimental research design.

In addition, we consider two constructs for the control group as do Salvanes et al. (2024): a) those who never engaged, and b) those who engaged in the future. In the latter case, as our panel is much shorter than theirs, to trade-off between comparability and analysis horizon, we employ a variant of it–individuals who are not yet treated–instead of a fixed time lag, and combine this setup with the CS DiD approach as an alternative robustness check. Yet, we acknowledge there might still be potential limitations to our identifying assumptions, particularly if individuals participated in social activities rather frequently throughout a survey year. In such cases, observing engagement behaviors would be certain rather than random. To alleviate this concern, we adopt a dynamic estimator to show no differences in studied outcomes between the treatment and control groups prior to the inception of social engagement. To increase the plausibility of potential randomness, the parallel trends assumption is assumed to hold conditional on a set of time-varying covariates that are essential to the paths of subjective survival (see Caetano et al., 2022; Roth et al., 2023). This assumption is further tested by allowing for covariate-specific trends addressed in Callaway and Sant'Anna (2021). Moreover, we construct an alternative treatment group by constraining the observations to those who reported their engagement was on an infrequent basis. In this case, comparisons are only made between those who occasionally participated in social activities and those who did not (or those who infrequently participated later). In sum, this paper includes two types of treatment groups and two types of control groups for each setting.

3.2 Models

The baseline two-way fixed-effects (TWFE) DiD model is as follows:

$$\text{SPUS}_{it} = \beta_1 \text{Engage}_{it} + \beta_2 X_{it} + \gamma_i + \gamma_t + \gamma_d + \epsilon_{it} \tag{1}$$

where SPUS_{it} is the subjective probability of unsuccessful survival of an individual i identified in wave t; Engage_{it} is a binary variable, equal to 1 if individual i participated in at least one social activity last month in wave t, and 0 otherwise; X_{it} is a row vector of individual characteristics, including age, its quadratic term, retirement status, and those discussed in Subsections 2.3 and 2.4; γ_i are individual fixed effects; γ_t are the survey wave fixed effects; γ_d are fixed effects of the distance between age at time t and the expected lifespan defined in CHARLS; ϵ_{it} is an idiosyncratic error term. Once an individual is treated in a wave, say 2015, yet changes to be untreated later on (2018 or 2020), we exclude those later periods from the estimation.[6]

The dynamic DiD model is as follows:

$$\text{SPUS}_{it} = \sum_{k=-3}^{-2} \beta_k \text{Engage}_{it}^k + \sum_{k=0}^{1} \beta_k \text{Engage}_{it}^k + \theta X_{it} + \gamma_i + \gamma_t + \gamma_d + \epsilon_{it} \tag{2}$$

where Engage_{it}^k denotes the status of social engagement of individual i at k periods relative to period t. k = -1 is excluded since the dynamic effect is compared to the period immediately before the treatment starts. The parameter of interest here is β_k, which estimates the effect k periods after or before the inception of social engagement. All other notations have the same meanings as in Equation (1).

As the timing for individuals to start their engagement differs, and utilities derived from such engagement may vary across individuals, we apply the CS DiD estimator, designed specifically for staggered DiD setups, to better address potential heterogeneity in treatment

[6] As mentioned earlier, considerable respondents exhibit an inconsistent engagement behavior. Hence, our strategy is to retain adequate observations for estimation while avoiding bias resulting from the switch in treatment status.

timing and effects. It also enables us to better assess the conditional parallel trends assumption, which is particularly suitable when covariate-specific trends in outcomes exist over time and when the distribution of covariates differs across groups.

The group-time average treatment effect (ATT) is defined as follows:

$$\text{ATT}(g,t) = E[\text{SPUS}_t(g) - \text{SPUS}t(0)|\text{Engage}_g = 1] \tag{3}$$

where $\text{SPUS}_t(0)$ denote the potential subjective probability of unsuccessful survival at wave t if individuals did not participate last month throughout the study period; $\text{SPUS}_t(g)$ capture the potential outcome if individuals were to be exposed to any social activities in period g for the first time. In other words, individuals will be sorted into different groups based on the wave in which they start engaging in social activities. It then estimates more aggregate causal parameters by summarizing the ATTs across different lengths of exposure, $e = t - g$. Consistent with the previous strategy, we exclude observations, when necessary, to forbid treated individuals to switch out of the treatment group.

The same estimation strategies are applied to the inverse scenario where we examine the causal effects of social disengagement on the SPUS, despite a nuance that we use respondents who consistently engaged in social activities across all the waves as the counterfactuals.

4 Results

4.1 Overview

Figure 1 summarizes the DiD coefficients, using Equation (1). Notably, controls with pre-treatment levels, as discussed in Subsection 2.4, are applied in Figure 1(a), whereas in Figure 1(b), they are allowed to vary throughout the study period for comparison. The results indicate that the inception of social engagement leads to a decrease in older people's SPUS by 2.04 to 2.08 percentage points, as shown in Figure 1(a), and by 1.98 to 2.26 percent-

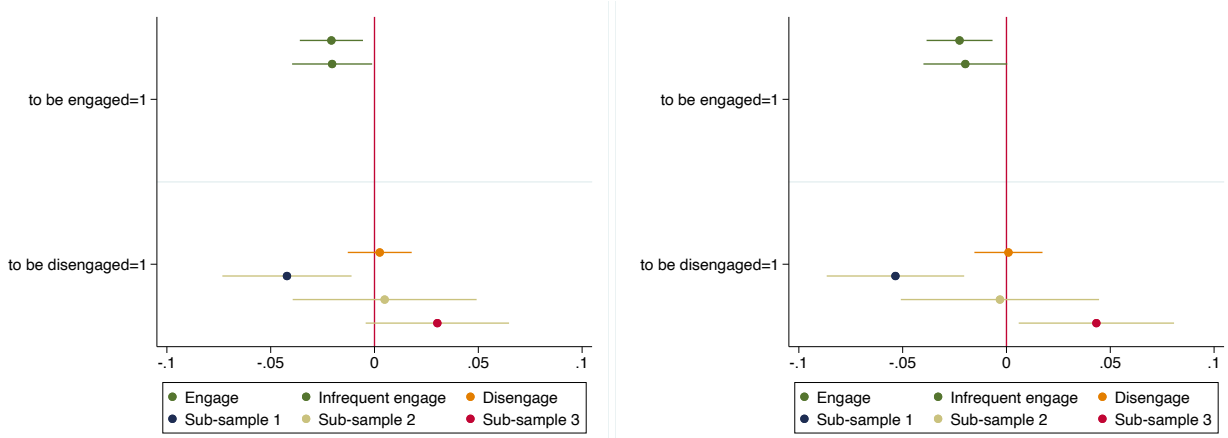
age points, as shown in Figure 1(b). In contrast, when individuals disengage from existing social activities, there appears to be no significant impact. For this reason, dynamic DiD estimation is not performed in this case. Additionally, although not reported here, retirement status in our case does not appear to alter either the engagement or disengagement effects on the SPUS, different from Gannon et al.'s (2014) finding that the association of social capital–measured through volunteering, caring, networking and family ties–and health diminishes at retirement.

Figure 2(a) plots the dynamic effects reported in Column (1) of Panel A in Table 2, estimated using Equation (2). In contrast, all controls are allowed to vary over time in Figure 2(b). The pre-trends from both specifications show no noticeable differences, supporting the validity of the parallel trends assumption. Further, in Column (2) of Panel A, we examine an alternative treatment group, i.e., those who reported their engagement on an infrequent basis only. As shown in Table 2, there is a decrease of 2.22 to 2.80 percentage points in older people's SPUS immediately after the engagement begins, followed by a decrease of up to 3.76 percentage points one period later, equivalent to approximately 8.3% of the sample mean. In contrast, none of the pre-treatment estimates are statistically significant. To further address the potential concern with selection bias, we break down the analysis with respect to mental health and current health status in the Supporting Information.

Furthermore, Figures 3(a) and 3(b) restrict the observations to more healthily homogeneous groups. Excluding all respondents who ever reported disabilities or serious diseases from the analysis would create the most ideal subsample. However, since this study focuses on older people, only 29% of them reported having no serious or chronic diseases. Therefore, we exclude respondents who reported a newly cured or largely relieved symptom from the analysis, along with disabled ones. The rationale is that when individuals' diseases are sufficiently relieved and under control, they are more likely to rejoin social activities, and at the same time, decrease their subjective probabilities of unsuccessful survival. As shown, the results confirm the parallel trends assumption as well and are quite similar to those in

Figures 2(a) and 2(b), although here the magnitudes of the effects are slightly smaller: 1.62 to 2.27 percentage points immediately, and 2.76 to 3.17 percentage points one period later.

Figure 1: Summary of DiD estimates of social (dis)engagement on older people's SPUS



(a) Aggregate estimates with controls of pre-treatment levels.

(b) Aggregate estimates with time-varying controls.

Notes: "Sub-sample 1" includes observations with simultaneously disengaged spouses. "Sub-sample 2" is confined to those whose spouses participated in only one social activity. "Sub-sample 3" includes single or widowed respondents and those whose spouses engaged in more than one activity. Individual age, retirement status, number of serious and chronic diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in both specifications. Other covariates take only pre-treatment values in Figure 1(a) but are allowed to fully vary over time in Figure 1(b). Source: Created by authors using CHARLS data.

4.2 Heterogeneity Analysis for Social Disengagement

It is somewhat puzzling that social engagement improves (subjective) health, as demonstrated by reduced SPUS, yet disengagement does not worsen it. Previous studies suggest that a spouse's health capital can directly influence the health of their partner (Wilson, 2002), and spousal altruism plays a significant role in the relationship between health status and happiness (De Mello and Tiongson, 2009). In this instance, spouses are likely to act as substitutes by providing emotional support and other benefits similar to those typically gained through social life, and consequently, help mitigate the negative effects caused by disengagement. As a result, spending more time with one's spouse helps individuals adapt to

the absence of social life.

According to the scatter plots and fitted lines in Figure SI1(b), individuals in the treatment group tend to have a spouse who was less engaged in social activities, and their SPUS was distinctly higher than that of the control group. Given that it is unclear how engagement behaviors of spouses sway the other's SPUS, despite the presence of correlations, it is useful to break down the analysis based on marital and/or spousal engagement status to further explore the plausible reasons for the insignificant results observed above. In fact, as reported in Figure 1 (see Sub-sample 1), a decrease of 4.22 percentage points is observed among individuals whose spouses were concurrently disengaged. This is probably because couples can spend more time together, and spouses are likely to provide better support if respondents do not benefit effectively from their participation in social activities.[7] This is especially true for certain types of social interactions, such as those related to hierarchical social capital (Kabayama et al., 2017), as these can even worsen mental health. Since this subsample accounts for nearly half of the observations in the disengagement setting, it can highly confound the estimate of the full sample.

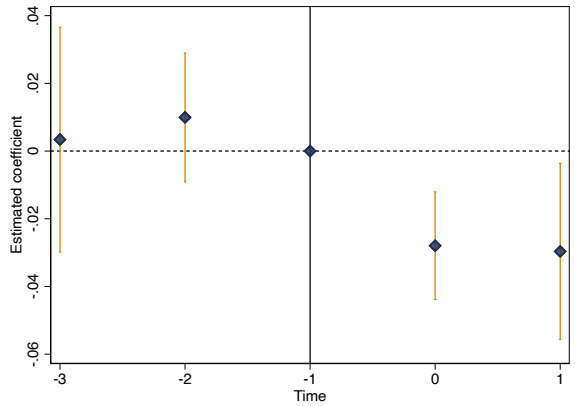
Interestingly, when respondents' spouses participated in only one social activity, no statistically significant increases or decreases in SPUS are observed in either specification (see Sub-sample 2). However, the effect becomes significant and positive when spouses participated in more activities. Moreover, disengagement most strongly elevates SPUS among single or widowed respondents, likely because these individuals are more prone to being alone once they stop participating in social activities. In other words, once individuals detach from social activities while their spouses still spend tangible time and effort socializing, or when they have no spouse, disengagement becomes more of a pure loss, as acquiring comparable value within households would be much harder.

[7]We consider such instances to reflect the low quality of social support received through social engagement compared to that provided by spouses. We do not believe that the decision to disengage is triggered by unfavorable participation or relationships, because if this were the case, simply exiting "toxic" surroundings would also create lower SPUS for respondents in other cases.

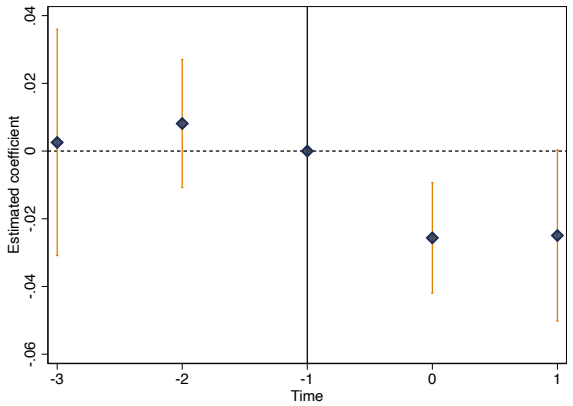
Since there are two subsamples exhibiting homogeneous responses, we proceed with estimation based on them. Figure 1 shows an average treatment effect of 3.03 and 4.33 percentage points from both specifications (see Sub-sample 3). We further report the estimates for each period in Columns (1)–(2) of Panel B in Table 2.[8] While no significant immediate effects are observed, older people's SPUS increases by 6.41 to 7.60 percentage points (14.2 to 16.8% of the sample mean) one period later. This is reasonable, as social ties formed through participation in social activities tend to fade gradually rather than disappear immediately. Additionally, dynamic effects using two sets of controls are presented in Figures 2(c) and 2(d). We then exclude disabled respondents and those who reported a newly diagnosed disease from the analysis, with the results reported in Figures 3(c) and 3(d). Together, these visual inspections reconfirm the parallel trends assumption and demonstrate an increase in older people's SPUS caused by disengagement.

[8]It should be noted that in Column (2), the treatment group is constructed by restricting observations to those with lower participation frequencies. While we apply a strict criterion in the engagement setting by focusing on individuals who reported infrequent engagement, here we relax the condition to include those who engaged on a weekly basis due to insufficient observations for further estimation. However, we believe this relaxation is not a major concern, as the counterfactuals are those who consistently engaged, rather than those who never engaged. Specifically, we compare individuals who disengaged after previously engaging on an infrequent or weekly basis with individuals who maintained their infrequent or weekly engagement patterns.

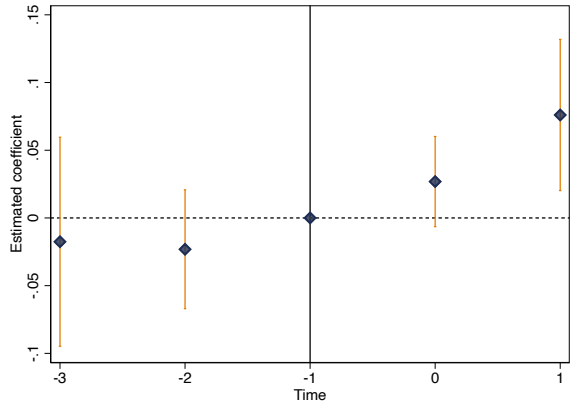
Figure 2: Dynamic effects of social engagement and disengagement on older people's SPUS



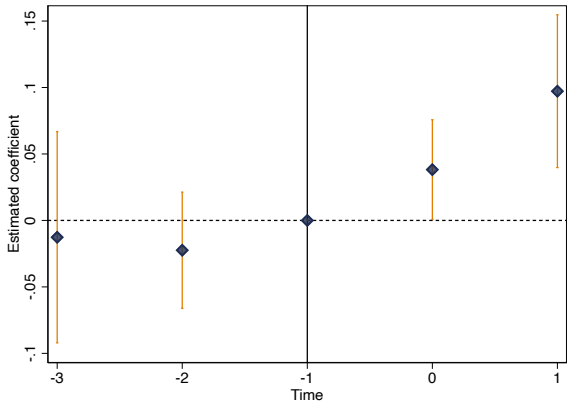
(a) Engaged versus not engaged with controls of pre-treatment levels.



(b) Engaged versus not engaged with time-varying controls.



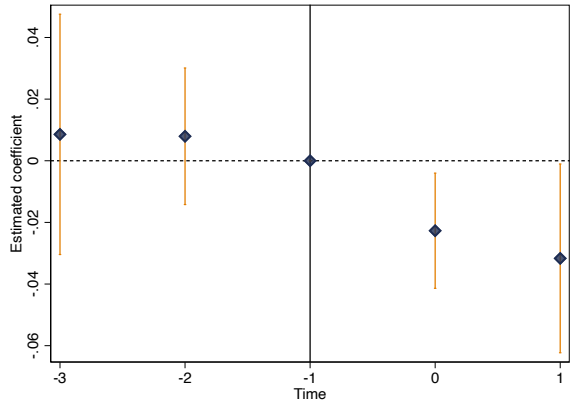
(c) Disengaged versus engaged with controls of pre-treatment levels.



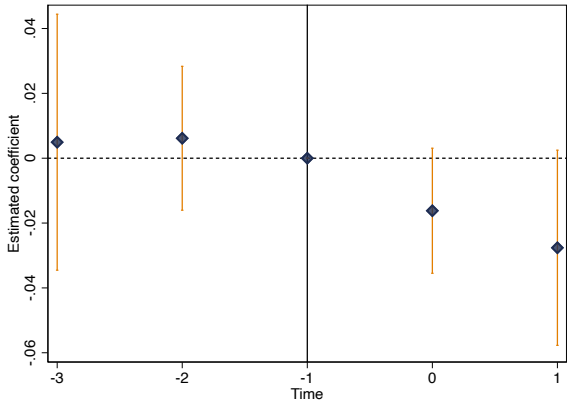
(d) Disengaged versus engaged with time-varying controls.

Notes: Estimates are presented with 95% CIs. R^2 is 0.6994 for Figure 2(a), 0.7189 for Figure 2(b), 0.7126 for Figure 2(c), and 0.7238 for Figure 2(d). Individual, time, and age-to-expected-lifespan distance fixed effects are used. Individual age, retirement status, number of serious and chronic diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values in Figures 2(a) and 2(c) but are allowed to fully vary over time in Figures 2(b) and 2(d).
Source: Created by authors using CHARLS data.

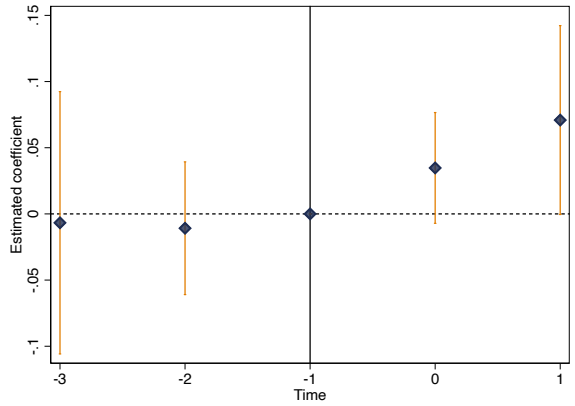
Figure 3: Dynamic effects of social engagement and disengagement on the SPUS of older people who do not experience related health shocks



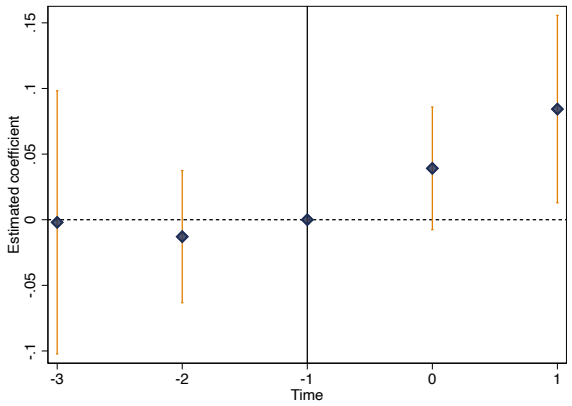
(a) Engaged versus not engaged with controls of pre-treatment levels.



(b) Engaged versus not engaged with time-varying controls.



(c) Disengaged versus engaged with controls of pre-treatment levels.



(d) Disengaged versus engaged with time-varying controls.

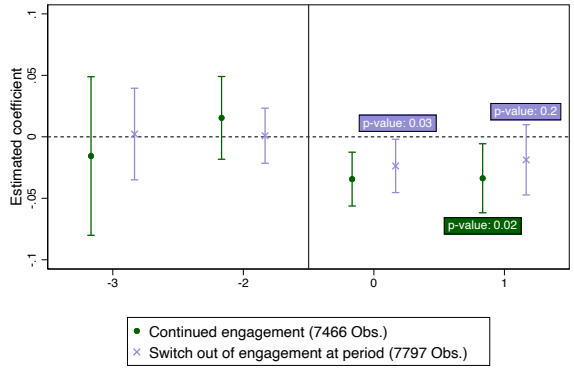
Notes: Estimates are presented with 95% CIs. R^2 is 0.7078 for Figure 3(a), 0.7261 for Figure 3(b), 0.7246 for Figure 3(c), and 0.7326 for Figure 3(d). Disabled respondents and those who reported a newly cured or largely relieved symptom are excluded from Figures 3(a) and 3(b). Similarly, disabled respondents and those who reported a newly diagnosed disease are excluded from Figures 3(c) and 3(d). Individual, time, and age-to-expected-lifespan distance fixed effects are used. Individual age, retirement status, number of serious and chronic diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values in Figures 3(a) and 3(c) but are allowed to fully vary over time in Figures 3(b) and 3(d).
Source: Created by authors using CHARLS data.

4.3 Comparisons Between Switchers and Non-Switchers

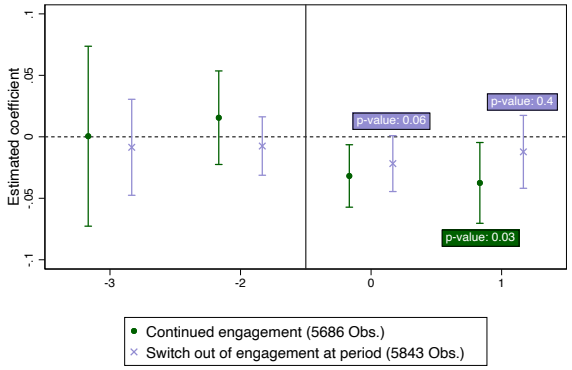
We acknowledge that selection bias may still arise under our current approach, as excluding observations that switch out of treatment later causes the composition of the treated group to shift across the treatment and post-treatment periods. In particular, this may result in post-treatment effects being driven by unobserved differences between respondents with continued (dis)engagement and untreated individuals, rather than by the treatment itself. Respondents who are consistently (dis)engaged could also be systematically different from those who switch in and out of (dis)engagement over time. To address this, we break down the treated individuals into two groups: (a) respondents who remained (dis)engaged in period 1, and (b) respondents who switched out of (dis)engagement in period 1. If the two groups exhibit similar patterns in period 0 but differ in period 1, with switchers "losing" the effects yet persistent effects among non-switchers, this would provide stronger evidence for identifying true treatment effects.

As shown in Figure 4, the two groups exhibit similar results in both settings in period 0, whereas the results begin to differ in period 1. Specifically, in Figure 4(a), switchers' engagement effects fade out and disengagement effects remain insignificant. In contrast, post-treatment effects among non-switchers remain significant. This asymmetry further supports the notion that disengagement effects take longer to materialize. Akin to Figure 3, the dynamic estimates in Figures 4(b) and 4(d) exclude respondents experiencing disease and disability shocks as discussed previously. The results remain robust and consistent with those in Figures 4(a) and 4(c). In sum, these findings suggest that (a) respondents with continued treatment and those who subsequently switched out of treatment are sufficiently comparable, and (b) frequent switching may attenuate treatment effects but it does not invalidate the effects identified in this study.

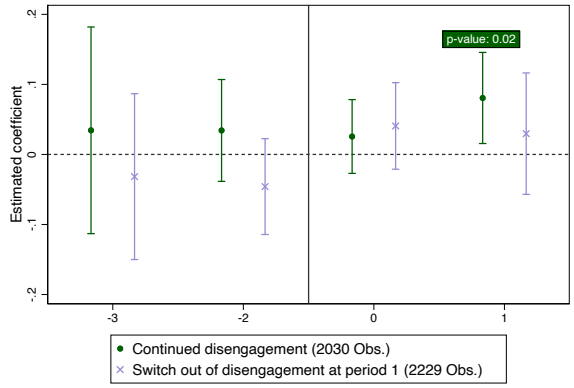
Figure 4: Dynamic effects comparisons: continued (dis)engagement versus switching out of (dis)engagement



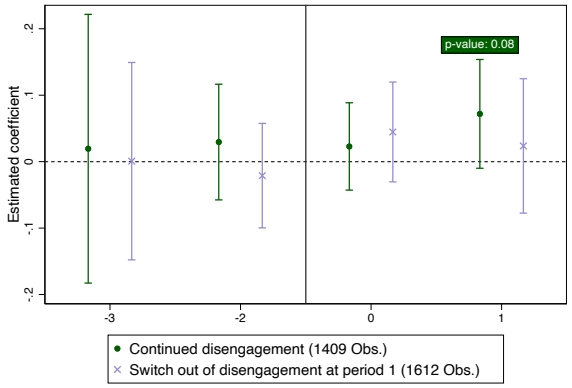
(a) Engagement effects comparisons.



(b) Engagement excluding health shocks.



(c) Disengagement effects comparisons.



(d) Disengagement excluding health shocks.

Notes: Estimates are presented with 95% CIs. Individual, time, and age-to-expected-lifespan distance fixed effects are used. Individual age, retirement status, number of serious and chronic diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values.
Source: Created by authors using CHARLS data.

Table 2: Dynamic DiD estimates for older people's SPUS

	(1) TWFE	(2) TWFE_IR	(3) CS	(4) CS_NY	(5) CS_IR	(6) CS_IR_NY
Time to Treatment						
Panel A w/ Treatment: Not engaged → Engaged						
-3	0.0034	0.0052	0.0132	-0.0273	0.0132	-0.0273
	(0.0169)	(0.0173)	(0.0203)	(0.0451)	(0.0203)	(0.0451)
-2	0.0010	0.0126	0.0157	0.0054	0.0157	0.0054
	(0.0097)	(0.0099)	(0.0111)	(0.0191)	(0.0111)	(0.0191)
-1	0	0	0	0	0	0
	0	0	0	0	0	0
0	-0.0280***	-0.0222**	-0.0270***	-0.0199*	-0.0193*	-0.0114
	(0.0081)	(0.0102)	(0.0095)	(0.0115)	(0.0116)	(0.0136)
1	-0.0296**	-0.0376**	-0.0486***	-0.0273	-0.0667***	-0.0462*
	(0.0133)	(0.0171)	(0.0169)	(0.0237)	(0.0215)	(0.0279)
Obs.	11,396	8,456	9,941	6,174	7,720	4,076
Panel B w/ Treatment: Engaged → Not engaged						
-3	-0.0176	0.0304	-0.0135	-0.0100	-0.0501	-0.0341
	(0.0393)	(0.0557)	(0.0568)	(0.0567)	(0.1082)	(0.1068)
-2	-0.0231	0.0061	-0.0270	-0.0198	-0.0166	-0.0151
	(0.0224)	(0.0306)	(0.0294)	(0.0290)	(0.0478)	(0.0461)
-1	0	0	0	0	0	0
	0	0	0	0	0	0
0	0.0269	0.0192	0.0310	0.0208	0.0035	0.0049
	(0.0170)	(0.0204)	(0.0256)	(0.0234)	(0.0317)	(0.0300)
1	0.0760***	0.0641**	0.0754*	0.0696[p:0.108]	0.1476***	0.1108**
	(0.0284)	(0.0326)	(0.0449)	(0.0434)	(0.0548)	(0.0505)
Obs.	2,814	1,733	2,190	2,304	1,264	1,328
Controls	Y	Y	Y	Y	Y	Y
Ind. FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Dist. FE	Y	Y	N	N	N	N

Notes: The DV is the subjective probability of unsuccessful survival. Columns (2), (5), and (6) in Panel A consider engagement behaviors on an infrequent basis. Columns (4) and (6) in Panel A use not-yet-treated individuals as the control group. Similarly, Columns (2), (5), and (6) in Panel B consider engagement behaviors on an infrequent or approximately weekly basis. Columns (4) and (6) in Panel B use a combination of never- and not-yet-treated observations as the control group. Individual age, retirement status, number of diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values. Individual, time, and age-to-expected-lifespan distance fixed effects are included in Columns (1)–(2), while the CS estimation does not allow for additional fixed effects. The CS estimates are generated using Callaway and Sant'Anna's (2021)'s DiD estimator with multiple periods and event study aggregation. Robust standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: Authors' elaboration using CHARLS data.

5 Robustness Checks

5.1 CS Estimator with Alternative Counterfactuals

The dynamic DiD results based on the CS estimator are reported in Columns (3)–(6) of Table 2, where Columns (4) and (6) consider not-yet-treated individuals as the control group, while the treatment group in Columns (5) and (6) is confined to observations with only infrequent engagement. Unfortunately, with the restrictions specified for Columns (4) and (6), we do not have enough observations in the disengagement setting to achieve relatively reliable estimation,[9] so we resort to a combination of never and not-yet-treated individuals as the counterfactuals.

Table 2 confirms the validity of the parallel trends assumption illustrated earlier for both the engagement and disengagement settings. It also shows larger magnitudes of the coefficients for the post-treatment period, suggesting that the TWFE models tend to underestimate the post-treatment effects and that greater effects are observed with increased comparability. Specifically, these results indicate that the onset of social engagement reduces older people's SPUS by 1.93 to 2.80 percentage points immediately and by 2.96 to 4.86 percentage points one period later. In contrast, while withdrawal from existing social activities does not appear to have an immediate effect on the SPUS, it results in an increase of SPUS by 6.41 to 14.76 percentage points one period later.

A caveat is that the estimates in the two scenarios are not fully comparable, as Panel B relies on only two subsamples, making it challenging to accurately compare the "rewards" of social engagement with the "penalties" of disengagement. Nevertheless, the larger post-treatment effects observed in Panel B compared to Panel A provide partial evidence of stronger disengagement effects, suggesting that social disengagement is likely to be more negatively influential to individual health than engagement effects are positively, though this

[9] Specifically, we can obtain only one estimate for one period in three and one group, respectively, using the comparable specifications in Columns (4) and (6).

impact occurs neither immediately nor universally.

5.2 Factor Analysis for Social Capital

Since the treatment variable in this study is constructed upon eight different types of social activities, it is likely that some activities generate social capital for participants, while others do not. Although it is difficult to clearly identify which activities act as "generators," social capital can be formulated as a latent factor derived from these eight activity measures. To capture this, we perform factor analysis to create an alternative treatment variable. Notably, this approach, by shifting the underlying assumption about the treatment from simple (dis)engagement to the gain or loss of social capital, is likely to produce larger effect magnitudes. Specifically, each activity is categorized as: a) not engaged, b) engaged on an infrequent basis, c) engaged on a weekly basis, and d) engaged on a daily basis, reflecting the idea that the degree of engagement in a single activity may also influence the generation of social capital. The factor analysis yields one principal component that explains almost the entire variance. We use a component score of 0.3, representing the 70th percentile, as the cut-off to classify whether respondents gained social capital through their participation.

In the engagement setting, respondents with a component score above 0.3 are assumed to have gained social capital and are classified into the treatment group, while those who are not socially engaged or less engaged serve as the counterfactuals. In the disengagement setting, respondents whose component scores are initially above 0.3 but who later become disengaged are classified into the treatment group. Conversely, respondents who maintain consistent engagement or are consistently engaged only to a moderate extent form the control group, respectively. The corresponding results using two sets of controls are reported in Columns (1)–(4) of Table 3. As shown, the results are generally consistent with our previous findings but present stronger immediate effects, compared to our baseline TWFE results. Interestingly, disengagement here is found to have an immediate effect on older

people's SPUS, and this effect is also evident when using the full sample. This suggests that, despite the gradual erosion of social ties following disengagement, a shift in social capital can promptly influence older people's health perceptions.

5.3 Ordinal Response Models and Monotonic Transformations

Ferrer-i Carbonell and Frijters (2004) demonstrates that accounting for time-invariant unobserved factors is more critical than the distinction between treating subjective measures like well-being as cardinal or ordinal, yet a fixed effects estimator can be significantly biased in non-linear models (Beck, 2020). However, the advantages of applying fixed effects in OLS regressions cannot address two concerns of note remained in this study: a) the distance between "not very likely" and "maybe" might not be the same as the distance between "very likely" and "almost certain," and b) the treatment effects (i.e., the signs of coefficients) from both OLS and ordinal regressions can be reversed when using ordinal outcomes (Schröder and Yitzhaki, 2017; Bond and Lang, 2019), although such reversals are only possible when respondents answer survey questions in a highly non-linear manner (Kaiser and Vendrik, 2020).

To address these concerns, we firstly conduct multilevel ordered logistic regressions to further examine the results, balancing the need to account for the ordinal nature of our DV and unobservable factors. The new findings are presented in Columns (5)–(8) of Table 3, where respondents are treated as the higher level to absorb unmeasured individual-specific factors affecting the SPUS. Overall, the results are consistent with our previous findings. Secondly, following recent empirical studies (e.g., Lindqvist et al., 2020; Atalay and Edwards, 2022), we visually assess whether our main TWFE estimates are robust to a wide range of smooth convex and concave monotonic transformations of the DV, as suggested by Bloem (2022). Figure SI4 illustrates that our estimates are not sensitive to monotonic increasing transformations of the DV, with no indication of sign reversals in either engagement or

disengagement setting.

Table 3: Dynamic DiD estimates based on factor analysis and multilevel ordered logistic regression

	(1) looseC_pre	(2) looseC_v	(3) strictC_pre	(4) strictC_v	(5) all_pre	(6) all_v	(7) infreq_pre	(8) infreq_v
	Factor Analysis				**Multilevel Ordered Logistic**			
Time to Treatment								
Panel A w/ Treatment: Not engaged → Engaged								
-3	0.0204	0.0229	0.0103	0.0163	-0.1705	-0.1703	-0.1637	-0.1621
	(0.0176)	(0.0148)	(0.0199)	(0.0205)	(0.1160)	(0.1172)	(0.1182)	(0.1183)
-2	0.0034	0.0076	-0.0027	0.0032	0.0313	0.0305	0.0456	0.0410
	(0.0110)	(0.0094)	(0.0133)	(0.0133)	(0.0720)	(0.0729)	(0.0734)	(0.0735)
-1	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
0	-0.0337***	-0.0275***	-0.0338***	-0.0374***	-0.2368***	-0.1707***	-0.2603***	-0.2622***
	(0.0087)	(0.0078)	(0.0096)	(0.0100)	(0.0561)	(0.0550)	(0.0742)	(0.0751)
1	-0.0340**	-0.0124	-0.0293*	-0.0251*	-0.2244***	-0.1002	-0.2463**	-0.1943*
	(0.0163)	(0.0130)	(0.0172)	(0.0149)	(0.0863)	(0.0847)	(0.1129)	(0.1171)
Obs.	13,897	16,375	11,270	11,216	13,020	13,042	10,602	10,301
Panel B w/ Treatment: Engaged → Not engaged								
-3	-0.0217	-0.0221	-0.0333	-0.0401	-0.0369	-0.0051	0.2914	0.3606
	(0.0268)	(0.0272)	(0.0327)	(0.0343)	(0.2076)	(0.2114)	(0.4481)	(0.4741)
-2	-0.0073	-0.0069	-0.0111	-0.0136	-0.1298	-0.1324	0.0703	0.0373
	(0.0175)	(0.0306)	(0.0199)	(0.0205)	(0.1389)	(0.1406)	(0.2812)	(0.2962)
-1	0	0	0	0	0	0	0	0
	0	0	0	0	0	0	0	0
0	0.0381**	0.0366**	0.0526***	0.0572***	0.2354**	0.2324**	0.2926**	0.2799*
	(0.0167)	(0.0183)	(0.0190)	(0.0216)	(0.1035)	(0.1113)	(0.1295)	(0.1434)
1	0.0373*	0.0426*	0.0702**	0.0851***	0.3343**	0.4115**	0.4038**	0.4415**
	(0.0212)	(0.0222)	(0.0283)	(0.0314)	(0.1795)	(0.1826)	(0.2047)	(0.2171)
Obs.	3,693	3,500	2,586	2,387	4,528	4,300	2,089	1,861
Controls	Y	Y	Y	Y	Y	Y	Y	Y
Ind. FE	Y	Y	Y	Y	N	N	N	N
Time FE	Y	Y	Y	Y	N	N	N	N
Dist. FE	Y	Y	Y	Y	N	N	N	N
Time trend	N	N	N	N	Y	Y	Y	Y

Notes: The DV is the subjective probability of unsuccessful survival. The counterfactuals in Columns (1)–(2) of Panel A consist of respondents who are either not socially engaged or engaged to a lesser extent, whereas those in Columns (3)–(4) of Panel A include only respondents who are not socially engaged. Similarly, the counterfactuals in Columns (1)–(2) of Panel B consists of all respondents who are consistently engaged, while those in Columns (3)–(4) of Panel B include respondents who remain engaged only to a moderate extent. Individual age, retirement status, number of diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values in Columns (1), (3), (5) and (7), while they are allowed to fully vary over time in Columns (2), (4), (6) and (8). In addition, Columns (7)–(8) in both Panels A and B consider engagement behaviors on an infrequent basis. Robust standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$
Source: Authors' elaboration using CHARLS data.

5.4 Propensity Score Matching

Comparability between the treatment and control groups is key to our identification. To further enhance it, we combine PSM and DiD, following a body of applied research (e.g., Caliendo and Kopeinig 2008; Song and Sun 2016; Chang et al. 2022), to further check the robustness of our findings. Specifically, we include variables such as gender, ethnicity, number of children, religion, education level, marital status, living in downtown areas, and Hukou status to perform one-to-one matching. The estimates conditional on the propensity scores are presented in Columns (1)–(2) of Table 4. Furthermore, we incorporate all the control variables used in the DiD specifications (taking pre-treatment values) into the matching process and relax the matching condition to allow for replacement. This adjustment is to balance unobserved factors with the number of matched observations. Accordingly, five-nearest neighbors matching with a caliper of 0.01 is applied.[10]

Since PSM was originally designed for cross-sectional data (Rosenbaum and Rubin, 1983), the inclusion of time-varying covariates can produce inconsistent matching results for different observations of the same respondent, leading to potential issues as discussed in King and Nielsen (2019). However, not incorporating time-varying factors or including post-treatment covariates may also introduce bias (e.g., Smith and Todd, 2005). To address these concerns while considering the staggered nature of our DiD design, we employ two strategies: a) matching based on the baseline period (the first observation) and b) matching that involves all study periods. As the common support after one-to-one matching is fully overlapped, we present the distribution of propensity scores after five-nearest neighbors matching in Figure SI5. The PSM-DiD estimates based on the two strategies are reported in Columns (3)–(4) and Columns (5)–(6) of Table 4, respectively, where the parallel trends assumption is also validated. In sum, the results reconfirm causality and, in general, reveal larger magnitudes of the effects. Since PSM helps improve comparability across observations,

[10]This caliper is much smaller than 0.2 of the standard deviation of the logit of the propensity score (0.15 in our case), a threshold widely used for choosing appropriate calipers (see, e.g., Austin, 2009).

the results here also suggests a potential downward bias in our main results.

Table 4: PSM-DiD estimates

	(1) oto_pre	(2) oto_v	(3) IntT_pre	(4) IntT_v	(5) allT_pre	(6) allT_v
	one-to-one		**five-nearest neighbor**			
Time to Treatment						
Panel A w/ Treatment: Not engaged → Engaged						
-3	-0.0105	-0.0160	0.0251	0.0176	-0.0114	-0.0136
	(0.0437)	(0.0438)	(0.0304)	(0.0301)	(0.0246)	(0.0245)
-2	0.0022	0.0088	0.0220	0.0199	0.0111	0.0056
	(0.0247)	(0.0240)	(0.0172)	(0.0170)	(0.0132)	(0.0131)
-1	0	0	0	0	0	0
	0	0	0	0	0	0
0	-0.0529**	-0.0460**	-0.0534***	-0.0528***	-0.0347***	-0.0360***
	(0.0227)	(0.0229)	(0.0162)	(0.0167)	(0.0119)	(0.0123)
1	-0.0788**	-0.0776**	-0.0556**	-0.0397[p:0.105]	-0.0329*	-0.0229
	(0.0345)	(0.0340)	(0.0248)	(0.0245)	(0.0180)	(0.0179)
Obs.	1,830	1,773	3,295	3,014	6,614	6,038
Panel B w/ Treatment: Engaged → Not engaged						
-3	-0.0220	-0.0738	0.0350	0.0094	-0.0319	-0.0327
	(0.1066)	(0.1034)	(0.0714)	(0.0701)	(0.0572)	(0.0568)
-2	0.0300	0.0023	-0.0284	-0.0416	-0.0294	-0.0320
	(0.0532)	(0.0560)	(0.0417)	(0.0416)	(0.0299)	(0.0300)
-1	0	0	0	0	0	0
	0	0	0	0	0	0
0	0.0108	0.0722	0.0419	0.0842**	0.0154	0.0399
	(0.0494)	(0.0592)	(0.0319)	(0.0366)	(0.0223)	(0.0258)
1	0.1535*	0.3325***	0.1154**	0.1947***	0.0802**	0.1352***
	(0.0899)	(0.0785)	(0.0552)	(0.0586)	(0.0370)	(0.0400)
Obs.	421	374	779	703	1,492	1,351
Controls	Y	Y	Y	Y	Y	Y
Ind. FE	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y
Dist. FE	Y	Y	Y	Y	Y	Y

Notes: The DV is the subjective probability of unsuccessful survival. The results in Columns (1)–(2) are based on one-to-one matching, while those in Columns (3)–(6) are based on five-nearest neighbors matching with a caliper of 0.01. In particular, Columns (3)–(4) use matching based on the initial period, while Columns (5)–(6) use matching that incorporates all study periods. Individual age, retirement status, number of diseases, number of disabilities and ADLs and IADLs are included as time-varying controls in all specifications. Other covariates take only pre-treatment values in Columns (1), (3), and (5), while they are allowed to fully vary over time in Columns (2), (4), and (6). In addition to these controls, gender, ethnicity, number of children, religion, education level, marital status, living in downtown areas, and Hukou have been included for matching. Robust standard errors are clustered at the individual level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: Authors' elaboration using CHARLS data.

6 Discussion and Concluding Remarks

As people's awareness of mortality risk improves, their personal estimates can more accurately reflect their actual likelihood of survival in the future. And not only will social behaviors form subjective beliefs, personal beliefs also impact health behaviors like smoking, drinking and driving, and accordingly, actual health in the long run (Sloan, 2024). Thus, understanding how subjective survival is shaped, particularly by changeable factors, will generate substantial value in facilitating healthy aging, and eventually contribute to the sustainable development of society, especially considering that educated, mature adults tend to be more important to economic growth than educated young ones (Marconi, 2018).

This paper explores the causal effects of social engagement vis-à-vis disengagement on older people's subjective probabilities of unsuccessful survival, using a panel of 11,412 respondents from the 2011–2020 China Health and Retirement Longitudinal Study. It leverages the potential randomness in the timing of social (dis)engagement to create a DiD design and employs a dynamic approach to validate the parallel trends assumption, conditional on pre-treatment characteristics. To reinforce the identification, it constructs alternative treatment and control groups, using individuals who engaged only on an infrequent basis and those who did not yet engage but would do so later. Several robustness checks have been conducted, including Callaway and Sant'Anna's (2021) DiD estimator designed specifically for staggered DiD setups, an alternative treatment variable based on factor analysis, multilevel ordered logistic regressions, monotonic transformations of the DV, and the combination of PSM and DiD estimation.

Our primary results demonstrate that a) social engagement reduces older people's SPUS by 1.93 to 2.80 percentage points (4.3 to 6.2% of the sample mean) immediately after it begins and by 2.96 to 6.67 percentage points (6.5 to 15.9% of the sample mean) one period later; b) social disengagement has no immediate effects, but its post-treatment impact can be as high as 14.76 percentage points (32.6% of the sample mean), reflecting a gradual,

rather than instant, erosion of social ties following the withdrawal. When breaking down respondents into those with continued (dis)engagement and those who subsequently switch out of treatment, we find that the engagement effects persist among non-switchers but fade out among switchers once switching begins; the disengagement effects emerge among non-switchers but remain insignificant among switchers. In addition, retirement status does not appear to alter either the engagement or disengagement effects on the SPUS. However, in the latter setting, no bad consequences are observed among those whose spouses are also socially inactive. This is probably because disengagement tends to be a pure loss game for single or widowed respondents and those with actively socializing spouses, since equivalent value within households would be much less obtainable. Overall, the results provide partial evidence of stronger disengagement effects, yet the effects are found to occur neither instantly nor universally.

To conclude, this study is among the first to use a quasi-experimental design to examine how social behaviors interplay with health, and to the best of our knowledge, the first to focus specifically on survival beliefs. Although a handful of previous studies touch upon a similar topic (e.g., Ross and Mirowsky, 2002; Anstey et al., 2002; Wu et al., 2015; Hoppmann et al., 2017; Prati, 2022), they differ fundamentally from this study in three main ways: they focus chiefly on demographic characteristics or socio-economic status as determinants of subjective survival; they treat social engagement as a control or moderator for subjective well-being; and they base their analysis on correlations within the context of Western societies, such as the US, Germany, and Australia. Furthermore, the existing literature has focused more on social isolation and loneliness, and occasionally on reduced engagement in specific activities (e.g., Kawai et al., 2020; Hajek and König, 2021; Ejiri et al., 2022; Nemitz, 2022; Etheridge and Spantig, 2022). It is thus the first to explore a reverse scenario in which individuals transition from participation in social activities to detachment from them.

In sum, our findings suggest that policymakers and health practitioners in aging communities, nursing homes, or similar settings can consider developing real opportunities that can

motivate older people to join or return to social engagement to promote healthy aging. In addition, given that our data make certain specifications less feasible in the disengagement setting, we recommend that future research explores longitudinal data from other developing countries to compare with our results, or conducts fieldwork for data collection to further study the consequences of disengagement.

Declaration of Interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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